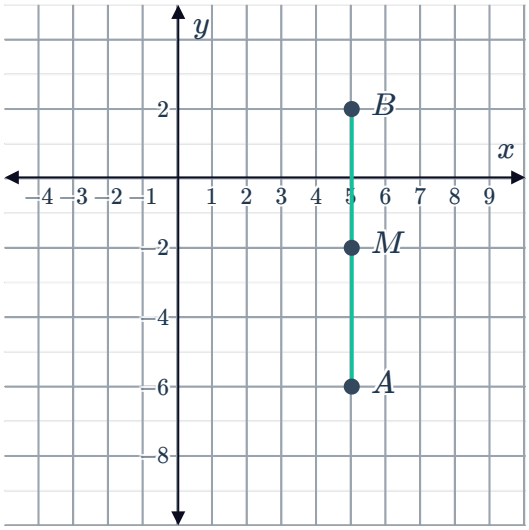
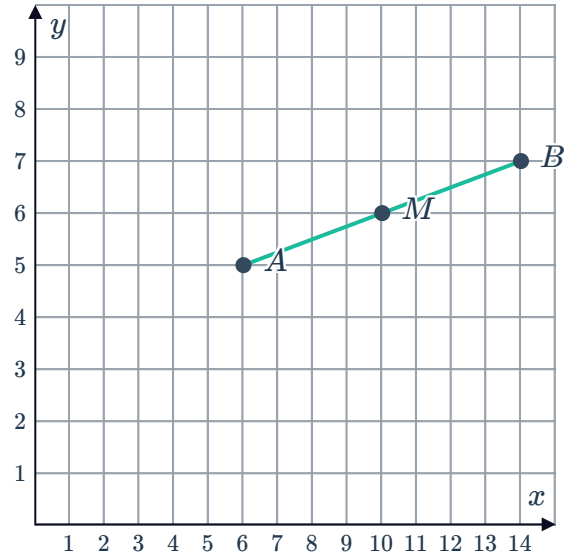
Midpoint of a segment

1 Consider the two points A and B with their midpoint M plotted on the number plane. Find the coordinates of M:

a $A\,(5,-6)$ and $B\,(5,2)$

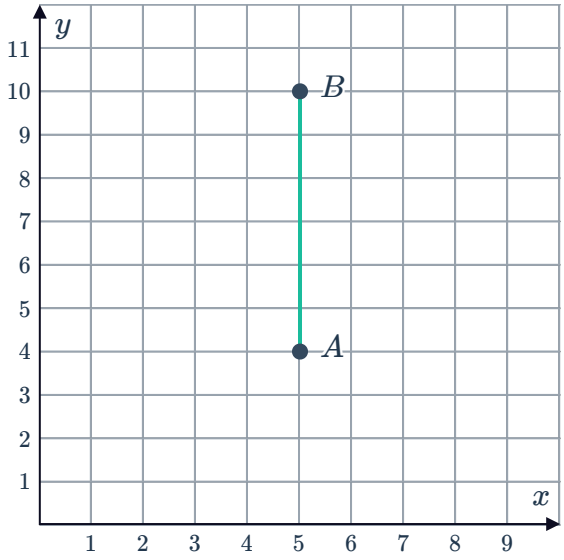


b $A\,(6,5)$ and $B\,(14,7)$

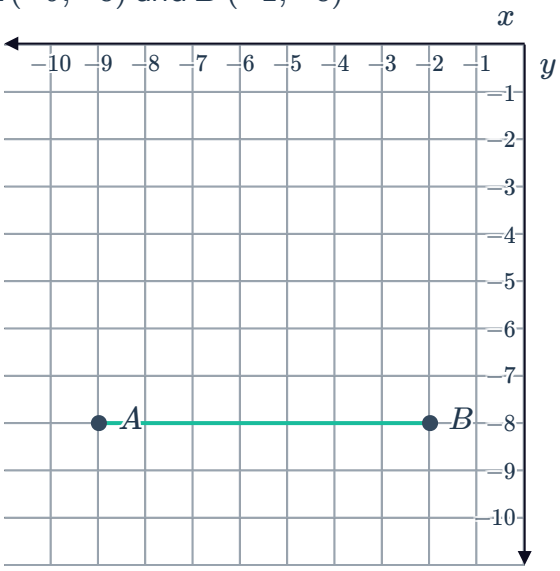


2 For each of the following find the coordinates of M, the midpoint of AB:

a $A\,(5,4)$ and $B\,(5,10)$

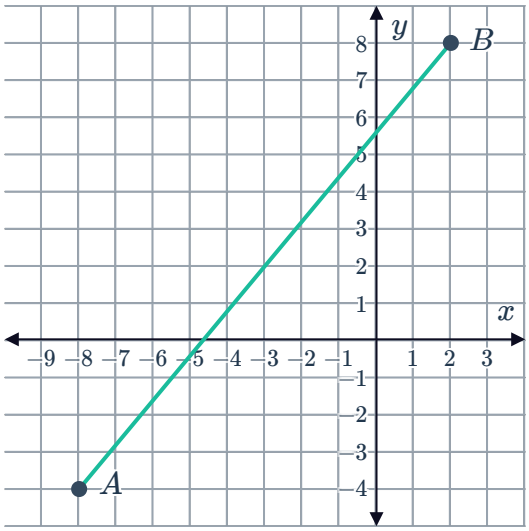


b $A\,(-9,-8)$ and $B\,(-2,-8)$

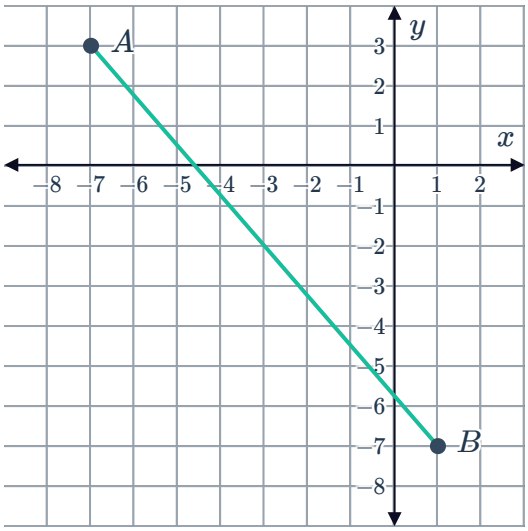


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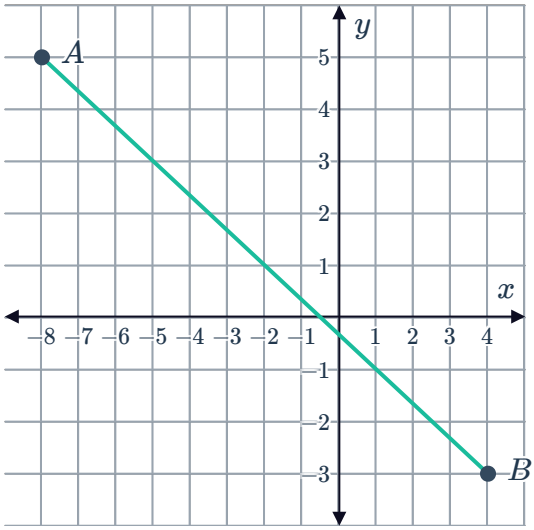
c $A\,(-8,-4)$ and $B\,(2,8)$



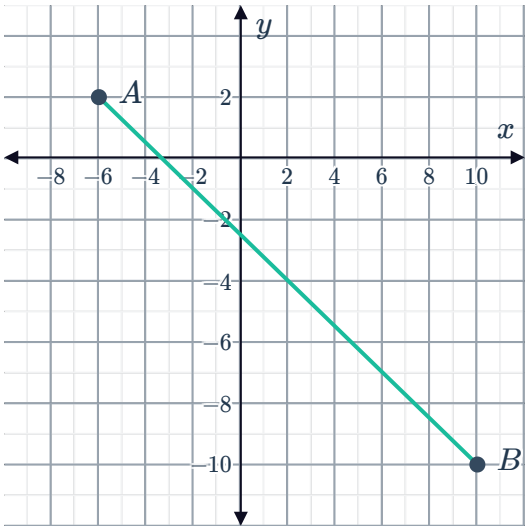
d $A\,(-7,3)$ and $B\,(1,-7)$



e $A\,(-8,5)$ and $B\,(4,-3)$



f $A\,(-6,2)$ and $B\,(10,-10)$



3 Consider the pair of points A and B given:

i Find the coordinates of the midpoint M.

ii Plot the segment AB and the point M on a number plane.

a $A\,(5,5)$ and $B\,(13,9)$

b $A\,(-12,-1)$ and $B\,(6,9)$

c $A\,(-9,1)$ and $B\,(5,-7)$

d $A\,(1,-11)$ and $B\,(9,7)$

e $A\,(4,5)$ and $B\,(6,9)$

f $A\,(-10,-1)$ and $B\,(4,5)$

g $A\,(-7,5)$ and $B\,(1,-9)$

h $A\,(-12,-6)$ and $B\,(4,-4)$

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4 Given that M is the midpoint of points A and B, find the coordinates of M in the following pairs of points:

a $A\,(7,3)$ and $B\,(9,9)$

b $A\,(5,7)$ and $B\,(2,-6)$

c $A\,(-9,8)$ and $B\,(-1,10)$

d $A\,(4,-6)$ and $B\,(8,-12)$

e $A\,(-11,-8)$ and $B\,(-7,0)$

f $A\,(-14,-4)$ and $B\,(4,8)$

g $A\,(-7,2)$ and $B\,(11,-14)$

h $A\,(-13,-12)$ and $B\,(5,-2)$

i $A\left(\frac{1}{2},-2\right)$ and $B\left(-2,\frac{7}{2}\right)$

j $A\left(\frac{7}{2},\frac{7}{2}\right)$ and $B\left(\frac{5}{2},\frac{3}{2}\right)$

5 Find the midpoint of $A\,(2m,5n)$ and $B\,(6m,n)$.

6 An interval AB has endpoints $A\left(9m-6,\frac{n}{6}\right)$ and $B\,(-7m,5n)$. Find the midpoint of the interval.

7 $M\,(4p+2,5q-3)$ is the midpoint of $S\,(20,-12)$ and $T\,(-18,6)$.

a Find the value of p.

b Find the value of q.

8 Consider the points $A(-10,10)$ and $B(16,4)$.

a Find the midpoint M of AB.

b For what value of k will the line $y=6x+k$ bisect the interval AB?

9 The points $PQRST$ lie on a straight line such that PQ, QR, RS and ST are equidistant. Find the points Q, R and S given $P\,(-4,-2)$ and $T\,(-12,10)$.

End points of a segment

10 Given that M is the midpoint of A and B, find the coordinates of A in the following given points:

a $B\,(9,6)$ and $M\,(7,6)$

b $B\,(10,-5)$ and $M\,(8,-5)$

c $B\,(9,12)$ and $M\,(7,9)$

d $B\,(11,2)$ is $M\,(8,-2)$

e $B\,(-1,0)$ and $M\,(-3,3)$

f $B\,(16,7)$ and $M\,(10,2)$

g $B\,(2,-9)$ and $M\,(-1,-7)$

h $B\,(-4,-6)$ and $M\,(1,8)$

i $B\,(-7,-9)$ and $M\,(-5,-3)$

j $B\,(-1,0)$ and $M\,(-4,-2)$

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11 Consider the given points A and M, where M is the midpoint of A and B.

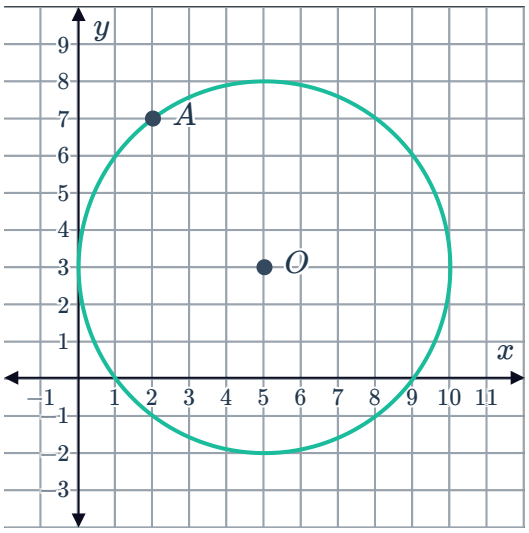
i Find the coordinates of point B.

ii Plot the segment AB and the point M on a number plane.

a $A\,(1,-1)$ and $M\,(7,7)$

b $A\,(-3,1)$ and $M\,(-4,-3)$

12 A circle with centre at Point $O\,(5,3)$ has AB as its diameter. The Point A has coordinates $(2,7)$ as shown:
Find the coordinates of B.



13 Consider the diagonals of a parallelogram.

a State the property of the diagonals of a parallelogram.

b $ABCD$ is a quadrilateral with vertices $A\,(-2,-1)$, $B\,(-10,3)$, $C\,(-15,7)$ and $D\,(-7,3)$. Find the midpoints of the diagonals.

i AC

ii BD

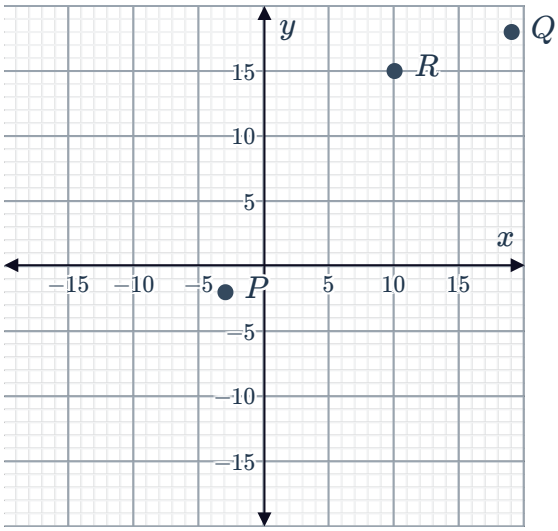
c State whether $ABCD$ is a parallelogram.

14 Consider the points $P\,(-3,-2)$, $Q\,(19,18)$ and $R\,(10,15)$.

a Find the coordinates of T the midpoint of PQ.

b Find the coordinates of S such that point T is the midpoint of RS.

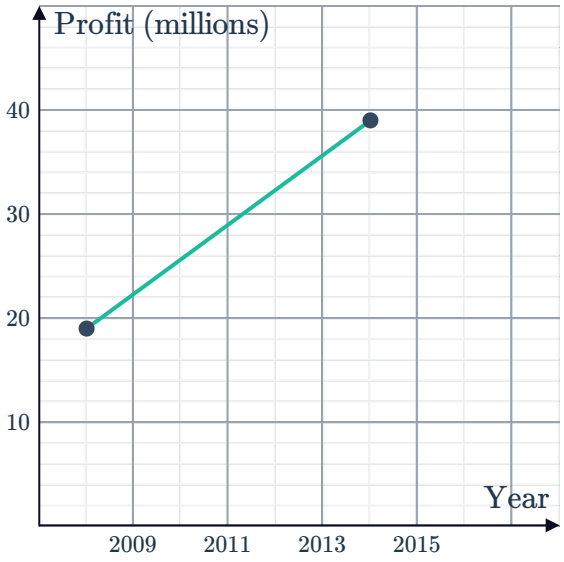
c What type of quadrilateral is $PRQS$? Explain your answer.



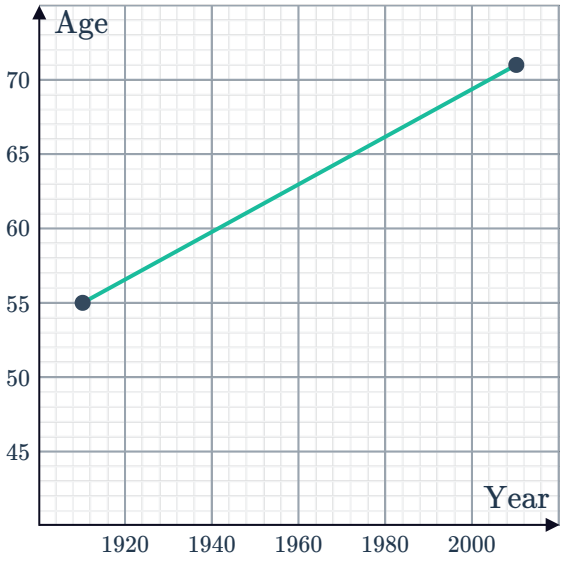
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Applications

15 The graph shows the annual net profit (in millions) of a company over the last few years. It shows that its profit has been growing approximately linearly from $\$19$ million in 2008 to $\$39$ million in 2014.
By finding the midpoint of the line segment, determine the company's net profit in 2011.



16 The graph shows a straight line that approximates the global life expectancy of a child over a period of 100 years. The graph shows the life expectancy to be 55 years in 1910, and 71 years in 2010.
Use the midpoint formula to estimate the life expectancy of a child in 1960.



17 Lines of latitude and longitude measure position on the Earth's surface and work like coordinates. The first coordinate represents how far above or below the equator you are, and the second coordinate measures how far from Greenwich Mean Time you are.
A plane starts its flight at $(20^\circ\text{N}, 62^\circ\text{E})$. It is bound for its destination at $(49^\circ\text{N}, 144^\circ\text{E})$. Find its position halfway through the flight.

18 The table shows the number of smartphone users in a particular city:

Year	Number of smartphone users
2007	175 904
2011	238 502
2015	298 080

a Using the midpoint formula, estimate the number of smartphone users in 2009.

b Using the midpoint formula, estimate the number of smartphone users in 2013.

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